MILITARY SPECIFICATION SHEET

METER, TIME TOTALIZING, MINIATURE,
DIGITAL, 5 VDC, SQUARE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirement for acquiring the meter described herein shall consist of this specification and the latest issue in effect of MIL-M-7793.

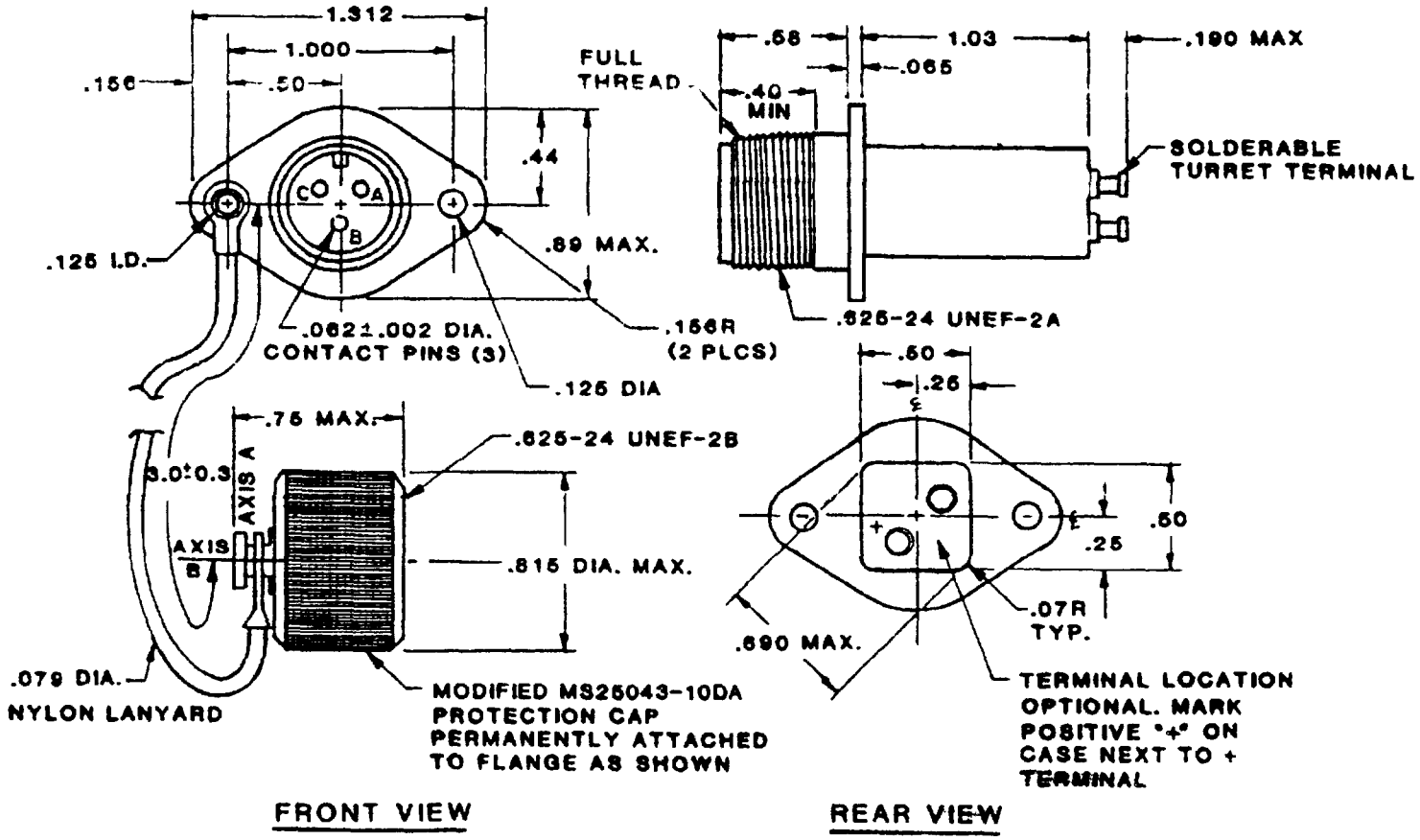


FIGURE 1.

AMSC N/A FSC 6645

DISTRIBUTION STATEMENT A. Approved for public release, distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T09:01Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

1. Dimensions are in inches. Unless otherwise specified, tolerances are ±.02 for two-place decimals and ±.015 for three-place decimals.

2. The mounting flange, threaded extension and threaded protection cap shall be constructed of aluminum alloy with an olive drab chromate coating over cadmium plating.

3. The protection cap shall be in accordance with MS25043-12DA except for the finish and the means of attachment. The finish shall be in accordance with Requirement 2 and the metallic chain shall be replaced with a nylon lanyard.

Performance:

Dielectric strength: The meter shall be capable of withstanding the application of 600 VRMS (room) and 350 VRMS (altitude) between the terminals and the case.

Life accuracy: ±0.1% from -65° to +125°C and 4.5 to 10 VDC.

Power consumption: Power consumption at 5 VDC shall not exceed 2 milliwatts.

Ripple voltage: Meter shall operate normally when subjected to a 2 volt peak (4 volts peak-to-peak) ripple between 10 Hz and 10 KHz superimposed on 7.0 VDC.

Operating temperature range: -65° to +125°C.

Part number: M7793/16-1.

Time range: Total readout of the meters shall be 99999.99 hours.

Meter output: The output from the meter shall be in a serial binary coded decimal format. The output impedance shall be 100K ± 10%. A logical "zero" shall be between 0.0 and +0.2 volts. A logical "one" shall be between +3.3 and +6.6 volts. The output pin assignments are shown in Figure 2 and the output data format is shown in Figure 3.

Transient protection: No temporary or permanent degradation shall be produced in the meter when subjected to ±25 volt transients of 10 microsecond duration occurring at a 1 millisecond repetition rate.

Durability of cap and threads: No mechanical or electrical degradation after 100 cycles in accordance with MIL-C-5015.

Voltage range: Operating voltage shall be from 4.5 to 10 VDC.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T09:01Z
Check the source to verify that this is the current version before use.

Durability of connectors and contacts: No mechanical or electrical degradation after 100 cycles in accordance with MIL-C-5015. Operational accuracy shall be checked at the end of the test.

Weight: Weight shall be not greater than 1.0 ounce with the attached protection cap.

Tensile strength of lanyard: The lanyard shall withstand a tensile load of 25 pounds when applied to Axis "A" and Axis "B" as shown on Figure 1.

Qualification tests: Qualification tests will be conducted in accordance with MIL-M-7793, except that the voltage, temperature and accuracy shall be as specified herein. Salt spray, moisture resistance, vibration, shock, electromagnetic compatibility and low temperature storage shall be conducted with the protective cap in place. Hermetic seal test is not applicable.

NOTES:

1. Elapsed time is read from the M7793/16-1 meter by mating the M7793/16-1 meter's connector with the M7793/12-1 reader's connector and operating the reader.
2. Digital readout and meter face plate per MIL-M-7793 are not applicable since the digital readout is provided by the separate M7793/12-1 reader.
3. The meter is designed to have an electrical ground and a continuous EMI/RFI shield with the threaded protection cap in place.
4. Contacts shall meet all the requirements of silver plated solder controls from MIL-C-5015.

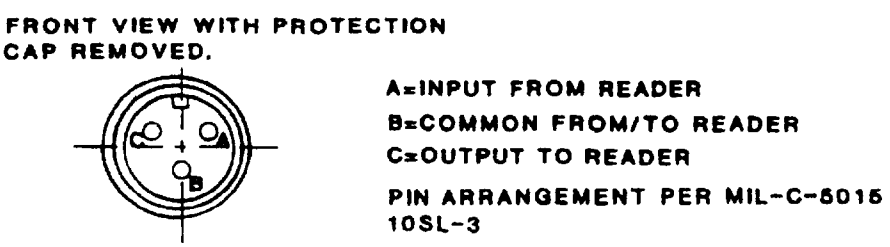


FIGURE 2. Readout pin assignment.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T09:01Z
Check the source to verify that this is the current version before use.

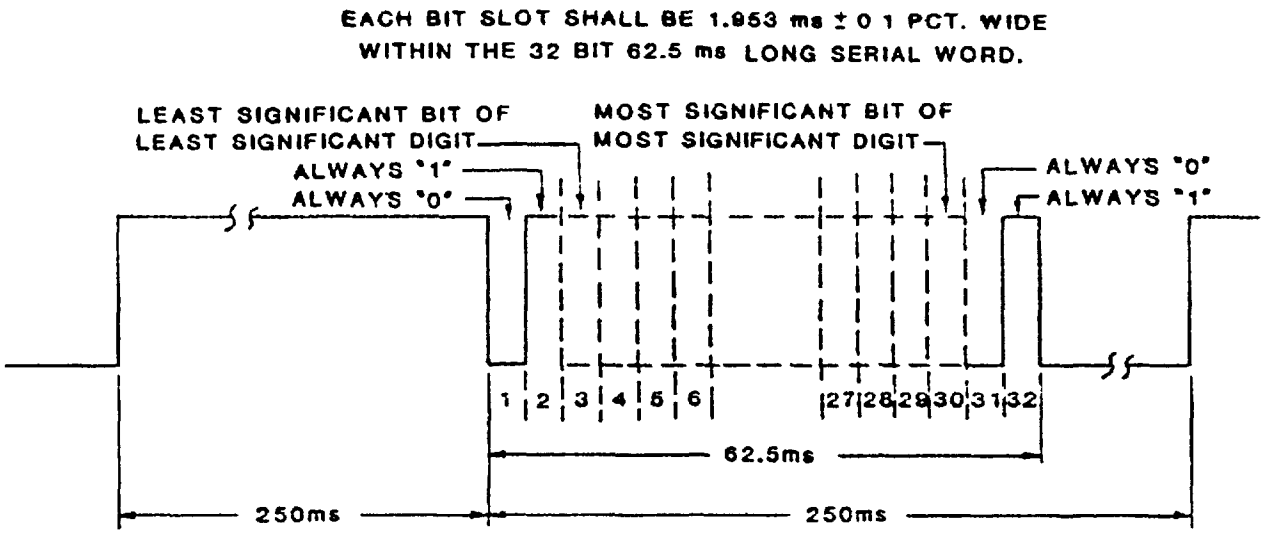


FIGURE 3. Output data format.

Custodians:
Navy - AS
Army -
Air Force - 99

Users:
Navy - SH, OS

Reviewer:
Army - MI

Preparing activity:
Navy - AS
(Project Number 6645-403)

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T09:01Z
Check the source to verify that this is the current version before use.